Administrative Continuity and State Capacity: Quasi-Natural Experiment Example from the Ottoman-Greek Divide

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Abstract

This paper examines whether the institutional boundary created by the 1832 Ottoman–Greek border produced persistent subnational differences in economic development. While much of the literature focuses on cross-national comparisons, we know far less about subnational variation in extractive state capacity in post-imperial contexts. Using the Arta–Volos line as a geographic regression discontinuity, I test whether municipalities that remained longer under Ottoman administration—and thus experienced Tanzimat and later reforms—exhibit stronger contemporary economic performance. Employing OLS and GRDD models with logged GDP (2021) and logged night-light luminosity (2020) as proxies for extractive capacity, I find no sharp discontinuity at the border. However, slope estimates consistently show that economic activity declines more steeply with distance on the Greek side, while the Ottoman side exhibits flatter and more favorable gradients. These results suggest that late Ottoman institutional reforms generated localized administrative advantages, qualifying standard negative interpretations of Ottoman legacy and highlighting the importance of reform depth and duration.

Note: Replication data and code are available upon request.

Introduction

A growing body of literature suggests that historical institutional arrangements can exert a long-lasting influence on contemporary economic outcomes (Acemoglu & Robinson, 2013; Popescu & Popa, 2022). While much of this work has centered on cross-national comparisons, there remains a significant gap in our understanding of subnational variation in extractive state capacity—particularly within post-imperial contexts (Harbers & Steele, 2020). This study seeks to address this gap by examining the first internationally recognized border between the Ottoman Empire and the newly established Kingdom of Greece—the so-called Arta-Volos line—formalized by the 1832 Treaty of Constantinople. I assess whether Greek municipalities that remained under Ottoman control for a significantly longer period exhibit higher levels of extractive capacity today compared to those incorporated into the Greek state shortly after independence.

The motivation for this inquiry builds directly on the methodological and conceptual framework developed by Vogler (2019), who uses a geographic regression discontinuity design (GRDD) to demonstrate that imperial bureaucratic institutions shape long-term bureaucratic efficiency and meritocratic hiring in Poland. However, whereas Vogler's study focuses on bureaucratic outcomes, this paper shifts attention to extractive state capacity—measured through contemporary economic performance—as a critical dimension of state-building and institutional endurance. Moreover, prior studies on the long-run consequences of Ottoman rule in Eastern Europe report predominantly negative effects on modern fiscal and financial development, operating through channels such as delayed human capital accumulation (Popescu & Popa, 2022) and constraints on formal banking associated with Islamic legal institutions (Grosjean, 2011).

These contributions, however, typically treat the Ottoman institutional environment as temporally uniform and devote limited attention to the nineteenth-century reform era—most notably the Tanzimat (1839–1876) and subsequent constitutional pe-

riods—which sought to introduce Western-style administrative, fiscal, and legal institutions. Whether these reforms generated subnational heterogeneity in institutional modernization remains largely unexplored. This paper aims to address this gap by examining whether exposure to late Ottoman reforms left a durable imprint on local extractive and administrative capacity. Replicating and adapting Vogler's GRDD strategy, I analyze whether exposure to Ottoman institutional reforms, particularly during the Tanzimat and constitutional periods, left a durable legacy on the local fiscal and administrative capacity in the regions they governed.

The Greek case presents a compelling natural experiment in institutional divergence. Following independence in 1832, the Kingdom of Greece emerged with weak central authority, nascent bureaucratic infrastructure, and limited administrative reach. State-building efforts were constrained by internal fragmentation, contested sovereignty, and reliance on foreign advisors. In contrast, municipalities that remained under the Ottoman rule—especially those in Western Thessaly and Thrace—continued to operate within an increasingly centralized imperial structure. During this period, the Ottoman Empire launched a series of deep reforms aimed at administrative modernization, the codification of law, the centralization of tax collection, and the professionalization of the military and bureaucracy. These institutional reforms likely contributed to the accumulation of administrative capacity and more routinized forms of extraction at the local level.

To identify the causal effect of these differing institutional exposures, I employ OLS and geographic RDD analysis centered on the 1832 border. By leveraging spatial proximity on either side of the former imperial boundary and controlling for confounders such as population density, distance to historical capitals and cities, and geographic controls, I predict the effect of Ottoman institutional persistence on economic outcomes—approximated by logged GDP in 2021 and VIIRS Annual Average night-light in 2020. The GRDD design ensures that municipalities near the border are otherwise

comparable in geography and culture, isolating institutional legacy as the key source of variation.

This paper makes two primary contributions. First, it applies a subnational GRDD framework to investigate long-term variation in economic development within a single modern state, highlighting how internal borders shaped by historical institutional breaks can produce persistent inequality. Second, it revises classical bellicist theories of state formation by showing that post-independence state-building is not merely shaped by conflict or resistance, but by the depth and continuity of pre-existing institutional infrastructures. In doing so, the paper contributes to a growing literature that recognizes how the quality and duration of prior statehood influence the uneven geography of economic development within modern nation-states.

Theoretical Framework

This study is grounded on the bellicist theories of state formation, which stresses the central role of coercion, extraction, and administrative consolidation in the development of effective states (Olson, 1993; Tilly, 2012, 2017). According to Tilly (2012, 2017), "War made the state and the state made war". Scholars of bellicist theories argue that as interstate wars became more costly, states were compelled to increase extraction, which contributed to the expansion of their capacity (Besley & Persson, 2009; Blaydes & Paik, 2016). This, in turn, spurred the development of other state institutions since taxation required information, coercion, monitoring, and related mechanisms. However, these theories have limitations in their explanatory scope, as they are primarily applicable to Western Europe (Stasavage, 2020).

Vogler (2019) explores how imperial bureaucratic rule shapes long-run bureacratic state capacity. Vogler uses historical imperial boundaries (e.g. Austrian-Prussian) in a quasi-natural experimental setting. Vogler's (2019) theoretical mechanism is based

on the expectation that imperial rule imposes its bureaucratic institutions, which persist as institutional legacies and determine contemporary bureaucratic efficiency and public service delivery. Vogler demonstrates that former Prussian territories in modern-day Poland, show a greater bureaucratic presence, and higher administrative capacity. Vogler's geographic RDD shows that territories once occupied by Prussia, maintain higher levels of bureaucratic capacity today.

Building on these insights, I argue that not only the imposition of imperial rule, as Vogler (2019) also showed the type of institutional organization matter, but also its duration and institutional depth is critical for post-independence trajectories. Ottoman municipalities that remained under imperial administration until the 1912 Balkan Wars experienced several waves of reforms, including the Tanzimat Era (1839-1876), First Constitutional Period, and Second Constitutional Period, unlike their counterparts in the newly independent Greek states that had to construct such state institutions from scratch. In addition, the Ottoman rule completely ceded Western Thrace and Macedonia to Greek control; it would have been fairly easier to integrate new territories into existing state structures and institutions, rather than consolidating institutions from the start. These advantages of Ottoman municipalities likely shaped their local state capacity and administrative structures.

Even though there are studies focused on the domestic aspects of state capacity, most of them employ agent-based model for their analysis (Levi, 1988; Besley & Persson, 2009). I aim to contribute to the literature, by leveraging OLS and geographic RDD models and shift the focus to institutional persistence in subnational level. In addition, my theoretical framework is built on the conceptualization of Berwick & Christia (2018). Previous literature focused on state as a homegonous entity, aimed at penetrating the society from a top-down view, however as Berwick & Christia (2018) stressed state-society dynamics are reciprocal. Thus, we conceptualize our theoretical expectation on institutional divergence between Ottoman Imperial rule and the newly independent

Greek state from that perspective. Late 19th century Ottoman reforms were crucial, especially in the Balkans, they transformed the demographics in the region. Furthermore, there are historical accounts that the nascent Greek state struggled to penetrate to local elites and had to build instiutions from scratch (Clogg, 2021; Koliopoulos & Veremis, 2009).

Previous literature highlights the long-term effects of imperial governance structures on contemporary state capacity (e.g. Acemoglu & Robinson, 2013; Vogler, 2019; Popescu & Popa, 2022; Grosjean, 2011). Drawing on this literature, my study examines how differences in the timing and nature of administrative integration into the Greek state, compared to the extended Ottoman rule, produced divergent legacies of extractive state capacity, potentially impacting economic outcomes. In particular, I argue that municipalities under longer Ottoman governance benefited from imperial reforms, which led to stronger administrative and extractive capacity that persisted to this day. Hence, my argument is that differences in post-independence extractive state capacity are not only a function of war, in contrast to the bellicist theories, or sovereignty, but of the duration and the efficiency of insitutional legacies.

The Ottoman–Greek Case

The Greek War of Independence, concluding with the Treaty of Constantinople in 1832, formalized the establishment of the new Greek state. The Treaty of Constantinople in 1832 defined the modern Greek state's initial borders (Arto-Volos line). This boundary not only separated the two states, but also imposed a sharp institutional divergence between the administrative units that would follow different institutional development.

On the Greek side, regions started constructing essential state institutions from scratch in the absence of prior institutional trajectories. As Clogg (2021) stated, these regions experienced a challenge of building state capacity without an inherited institu-

tional apparatus. Local elites dominated in the absence of functioning institutions which lacked tax capacity with leaving institutional voids (Koliopoulos & Veremis, 2009). By contrast, the Ottoman side, north of the 1832 line, remained under imperial rule for approximately 80 additional years (except for Thessaly), until the Balkan Wars of 1812. During this period, the Ottoman State implemented far-reaching administrative modernization reforms, most notably the Tanzimat, and the First and Second constitutional eras, which aimed at centralization, legal rationalization, and bureaucratic efficiency. As a result, regions under the Ottoman jurisdiction were exposed to evolving imperial administrative institutions over a much longer period.

The different institutional paths of the newly independent Greek state and Ottoman Empire motivated me to ask: Did exposure to Ottoman administrative reforms lead to higher local extractive state capacity and improve economic outcomes in contemporary Greek municipalities? Thus, my hypothesis is as follows:

H_1: Ottoman municipalities exhibit higher extractive state capacity as opposed to Greek municipalities around the cutpoint (as measured by log(GDP) and log(NTL)).

Replication of Vogler (2019)

Vogler (2019) examines how imperial administrative boundaries and bureaucratic legacies have long-term consequences for bureaucratic efficiency and meritocracy. Focusing on Poland, partitioned historically among the Russian, Prussian, and Austro-Hungarian Empires, he uses a geographic regression discontinuity design (GRDD) to identify persistent effects of imperial governance styles. He exploits historical border discontinuities and demonstrates that administrative units formerly under the Prussian rule exhibit significantly higher levels of bureaucratic efficiency and merit-based recruitment today. By contrast, regions historically governed by the Russian Empire score lowest on these indicators. His design leverages the quasi-random nature of imperial partitions near

the borders, assuming units in close proximity are comparable in all but institutional exposure. Vogler (2019) collected commune-level survey data as measures of both the dependent variables (e.g., meritocracy scores) and key covariates, and used logarithmic transformations of controls in his OLS models to reduce the influence of potential confounders.

My replication adopts Vogler's GRDD strategy but shifts the theoretical focus on imperial heterogeneity to post-sovereignty rupture. Instead of comparing rival colonial institutions, I examine the long-run impact of foundational discontinuities following independence. Specifically, the Greek-Ottoman border formalized by the 1832 Treaty of Constantinople established the territorial limits of the nascent Greek state. Unlike the Ottoman territories that remained under imperial rule until 1912 (with the exception of Thessaly, annexed to Greece in 1881), these early Greek municipalities had to construct new state institutions from scratch (Clogg, 2021). In contrast, Ottoman municipalities were subject to a highly centralized, evolving bureaucratic apparatus—especially following the Tanzimat reforms and the First Constitutional era—offering an additional 80 years of imperial institutional exposure.

This setup provides a compelling counterfactual to Vogler's (2019) analysis. While his design captures institutional differences across history, my study tests whether institutional maturity under continued imperial governance enhances extractive state capacity, even after eventual incorporation into the nation-state. Greek municipalities had inherited fragile and nascent institutions, but the post-1832 state gradually expanded to absorb new regions into an already functioning state structure. In contrast, the Ottoman side maintained administrative continuity and accumulated institutional experience until their later incorporation after 1912, potentially yielding higher long-term extractive capacity.

To maintain comparability with Vogler's (2019) design, I replicate his methodology: I implement an OLS analysis and supplement it with a GRDD approach using the

Imbens–Kalyanaraman (2012) bandwidth selection algorithm and reproduce his border plots and parametric specifications for robustness checks with varying bandwidths. While Vogler's treatment identifies variation in bureaucratic legacies shaped by colonial institutions, my treatment isolates the effect of a disruption in sovereign authority and the temporal scope of institutional exposure to centralized rule.

To further align with Vogler's study, I reconstructed his graphical design using employment-related data across historical borders. The figures below are taken from his replication dataset and illustrate the GRDD logic. Figure 1, for instance, plots the (logged) applications per job posting across communes near the Prussian-Russian border, showing a clear and substantial discontinuity—with communes on the Prussian side demonstrating higher administrative performance. This serves as a benchmark for visual and methodological comparison with my Ottoman-Greek application.

Figure 1: Prussia/Russia Comparison: Applicants/Job advertisement (Log.)

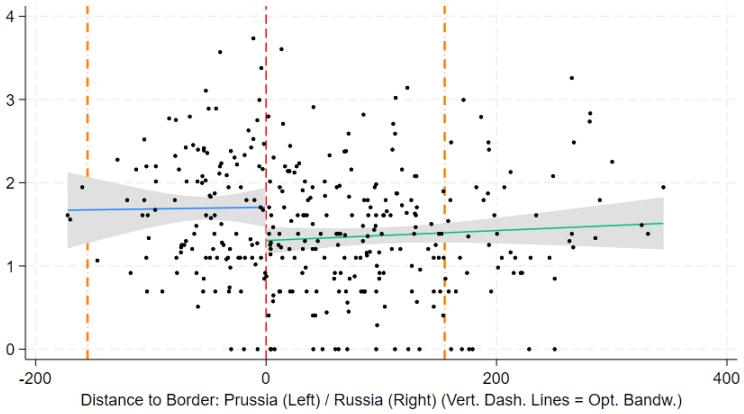
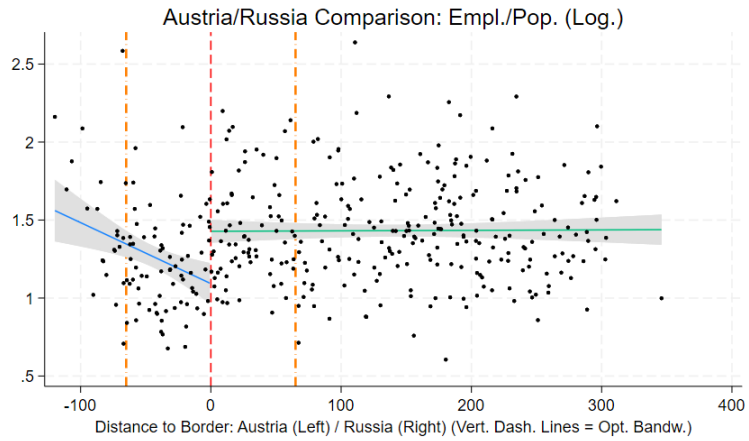


Figure 2, on the other hand, compares communes in their scopes on a different dependent variable and near the Austrian-Russian border. This time, Vogler uses the bureaucrat per capita variable for measuring bureaucratic efficiency.

Figure 2: Austria/Russia Comparison: Number of Employed bureaucrat/Population(Log.)



Both figures support his findings: Figure 1 shows that higher applicants per job on the Prussian side, imply higher meritocracy, whereas in Figure 2 more bureaucrat per capita on the Russian side near the border, implies inefficient bureaucracy.

Research Design and Data

In this study, I have estimated an OLS regression to estimate the effect of my treatment on modern economic outcomes. I have utilized the geographic RDD methodology that replicates Vogler's (2019) work. I shifted the focus of Vogler's paper from bureaucratic efficiency to economic outcomes of imperial legacies. I used the GRDD code of Vogler to replicate the same method in the case of the Greece-Ottoman border of 1832, and extend the study to include various geographic controls and robustness checks. My treatment is a binary variable indicating whether a municipality lies within the Ottoman side of the Arto-Volos line or on the Greek side. The construction of my dummy variable can be seen in the Equation . In order to construct my treatment variable, I have used a map created by Wilhelm Fischer (Fischer & Streit, 1837) dated 1837. The historical map can be seen in Figure 3. The pink-colored territories correspond to the Ottoman

Empire, whereas green refers to the newly independent Greek state. With the help of the QGIS software, I have constructed multipolygon and multiline geocoded data for the coordinates of municipalities and the Arto-Volos line. Later on, I have calculated the distance of each municipality to the border. I used centroids since the border intersects with municipality polygons, resulting in ineffective RDD findings.

$$\text{Post-1832}_i = \begin{cases} 1, & \textbf{Greek} \\ 0, & \textbf{Ottoman} \end{cases}$$

I have excluded the Aegean islands from the full sample, since most of the islands are locations of tourist attraction today, their GDP and NTL values would be inaccurate for my treatment. Hence, the labelled map is shown in Figure 4.

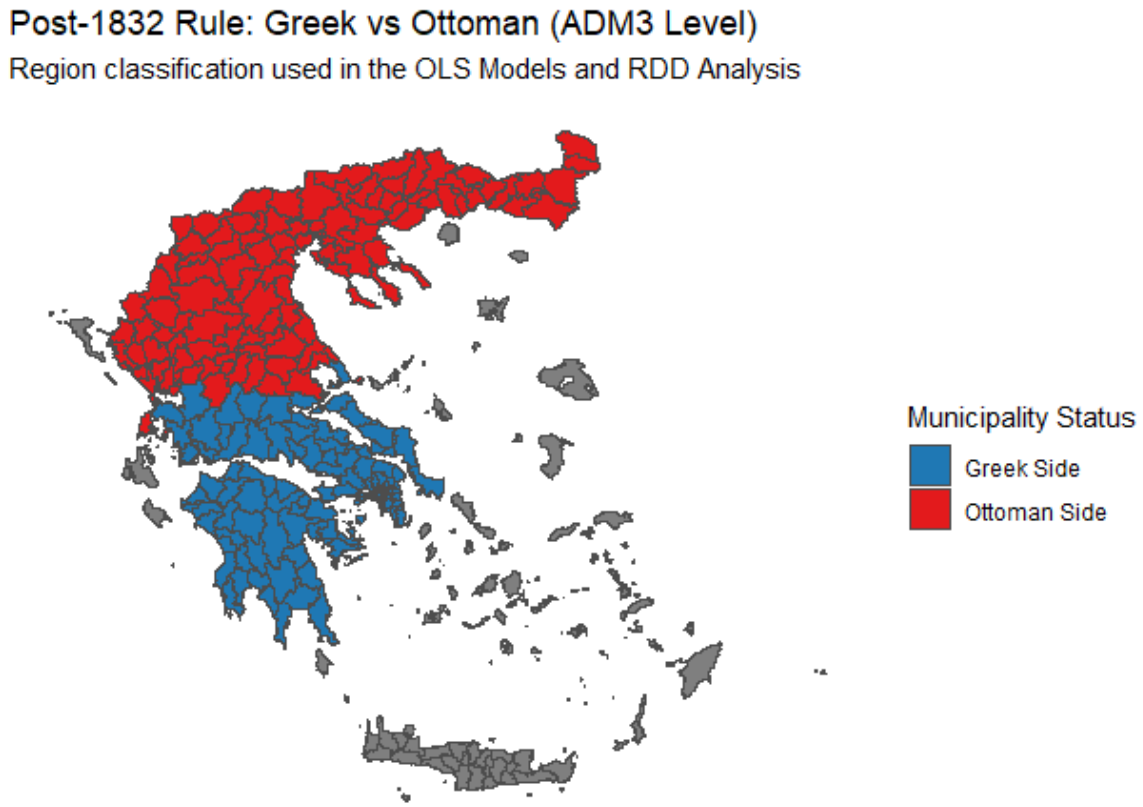
After constructing my treatment variable, I collected data for Greek municipalities. I have utilized the AidData database for retrieving data on Greece (Goodman, BenYishay, Lv, & Runfola, 2019). I retrieved geocoded data for ADM3 level units (corresponding to local municipalities in Greece) in multi-polygon format (Runfola et al., 2020). In order to measure extractive state capacity, I decided to use VIIRS average annual night-light and GDP as a proxy. I collected GDP per capita values of NUTS 2 regions in 2021 (Hellenic Statistical Authority, 2020), and then multiplied them with the population count from the urban population census (City Population, 2024) to calculate total GDP for ADM 3 level. I used logged GDP measures as a proxy for economic outcomes. Moreover, I collected VIIRS annual average nightlight and logged it to have an alternative proxy for the economic outcomes (Elvidge, Zhizhin, Ghosh, & Hsu, 2021). Furthermore, for our control variables, I retrieved geographic control variables (average temperature and precipitation for 1971-2000, distance to nearest coast, distance to nearest port, terrain ruggedness index), the area of the municipality and logged population density (Goodman et al., 2019). In addition, I have utilized the OSM (Open Street Map) (Mark Padgham, Bob Rudis, Robin Lovelace, & Maëlle Salmon, 2017) library in RStudio for collecting

Figure 3: Digitized historical GIS map, published by W. Natorff und Comp. Originally drawn for the “Atlas von Europa in zwei und achtzig Blättern”



data on highways and urban roads in Greece, *terra* library for computing the terrain ruggedness index.

Figure 4: Map of Greek Municipalities According to the Post-1832 Rule



After the data collection process has ended, in our secondary data generation process, I have constructed the final format of our data model. Additionally, I have calculated the area of each municipality using its multi-polygon coordinates. Also, I have taken the natural logarithm of the total population for each municipality, calculated the population density by dividing it by the calculated site in km^2 of the municipality. Finally, by simply multiplying GDP_{percap} x population to calculate GDP in the ADM 3 level.

My dataset includes 326 municipalities in total, but for accuracy, I have excluded islands. Since the Greek Riviera is one of the best regions for summer tourism, it could affect my findings. Hence, my final dataset consists of 248 observations. Each unit of analysis, namely i, corresponds to a municipality at the ADM 3 level in Greece. $Post_1832$ refers to our dummy variable, and *Distance* correspond to distance to the Arta-Volos line of the ith municipality centroid. *Distance* variable is signed distance to the cutpoint in km (positive for Ottoman municipalities, negative for Greek).

I estimated the same OLS model for both $log(GDP)$ and $log(NTL)$ dependent variables. My geographic RDD model is as follows:

Model 1:

$$\log(\text{GDP})_i = \beta_0 + \beta_1 \text{Post1832}_i + \beta_2 \text{Distance}_i + \beta_3 (\text{Post1832}_i \times \text{Distance}_i) + \epsilon_i$$

In this model, we predict log(GDP), Post1832 is our dummy variable indicating Ottoman and Greek municipalities, the closest distance to the border, and we included our treatment as the differential slope.

Model 2:

$$\log(\text{NTL})_i = \beta_0 + \beta_1 \text{Post1832}_i + \beta_2 \text{Distance}_i + \beta_3 (\text{Post1832}_i \times \text{Distance}_i) + \epsilon_i$$

We predict the same model on the logged annual average night-light in 2020, $log(NTL)$.

In addition, we also estimate the same models with the controls. In the models below, the term $\mathbf{X}_i'$ refers to the matrix of my controls. I include these covariates to account for any possible heterogeneity. It corresponds to, distance to the nearest port, the distance to the nearest coast, the distance to the historical capitals and cities, the area of municipality, the average temperature and precipitation, and the terrain ruggedness index.

Model 3:

$$\log(\text{GDP})_i = \beta_0 + \beta_1 \text{Post1832}_i + \beta_2 \text{Distance}_i + \beta_3(\text{Post1832}_i \times \text{Distance}_i) + \mathbf{X}_i'\boldsymbol{\gamma} + \epsilon_i$$

Model 4:

$$\log(\text{NTL})_i = \beta_0 + \beta_1 \text{Post1832}_i + \beta_2 \text{Distance}_i + \beta_3(\text{Post1832}_i \times \text{Distance}_i) + \mathbf{X}_i'\boldsymbol{\gamma} + \epsilon_i$$

Before proceeding to the OLS results, simple linear regressions can be observed in Figure 5 and 6, on log(GDP) and log(NTL) dependent variables, respectively. Gray-shaded areas correspond to the optimal bandwidth (Imbens & Kalyanaraman, 2012) which minimizes the root mean squared error of the model.

Figure 5: Income level of Municipalities around the Arto-Volos Line

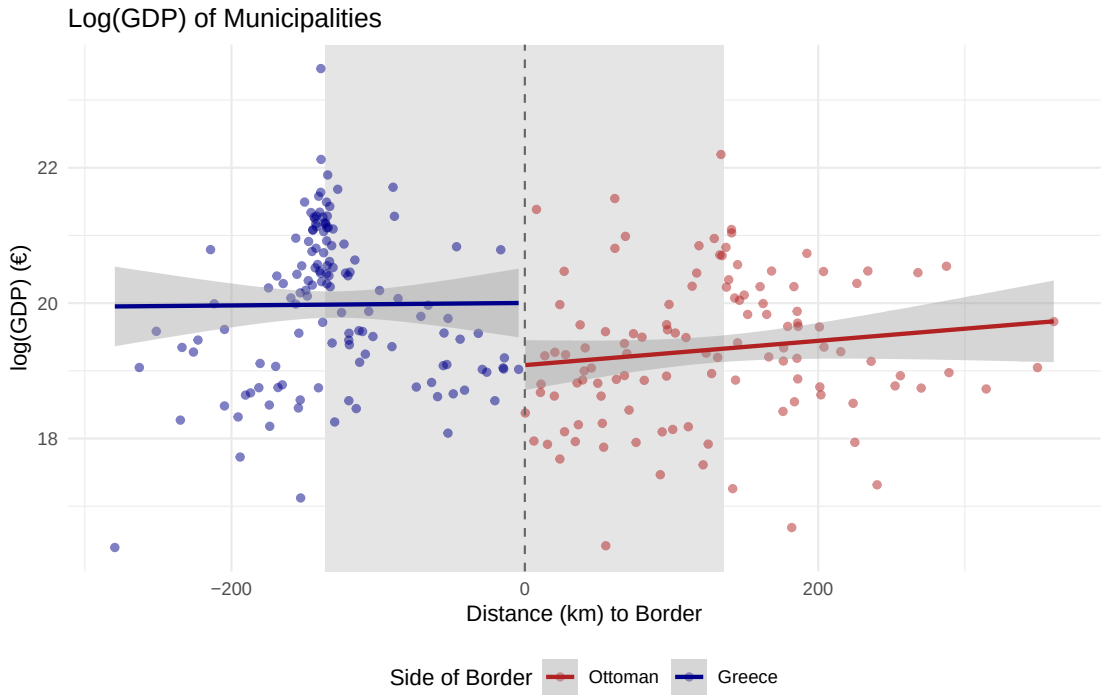
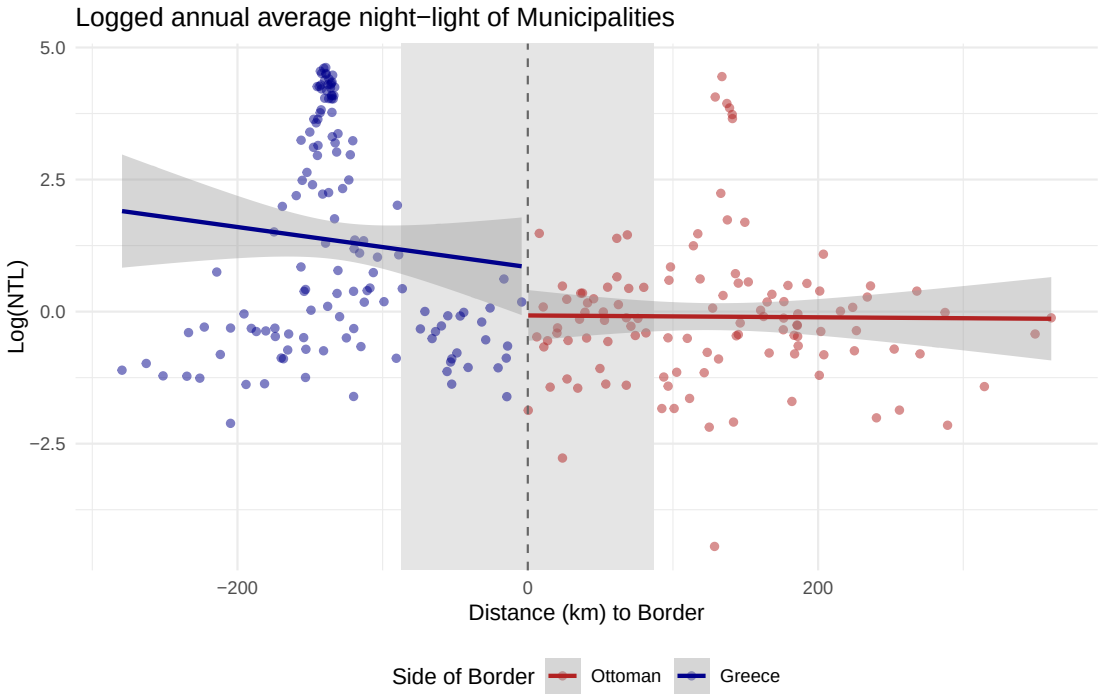


Figure 6: Socio-economic development in 2020 of Municipalities around the Border



OLS Estimates

Our OLS model estimates from Models 1 and 2 are reported in Table 1. While the additive models (Columns 1–2) estimate average level differences between Ottoman and Greek municipalities, they implicitly assume that the effect of distance from the 1832 border is identical on both sides. However, the historical argument motivating this study explicitly predicts that increasing distance should have different slopes depending on whether a municipality remained under Greek administration after 1832 or continued to experience late Ottoman reforms. Therefore, in Columns 3–4, I include an interaction between Post-1832 status and distance to allow the economic gradient to vary across institutional regimes. This specification allows me to test whether longer exposure to Ottoman administrative reforms produced a more gradual decay of economic activity compared to areas incorporated earlier into the Greek state. In both models,

our interaction term is not statistically significant.

Table 1: Baseline Models

	Dependent variable:			
	log(GDP) GDP Model	log(NTL) NTL Model	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.001	−0.001	0.002	−0.0002
	(0.001)	(0.002)	(0.001)	(0.002)
Post-1832 (Greek = 1)	0.989***	1.075**	0.918***	0.915*
	(0.305)	(0.494)	(0.321)	(0.520)
Distance x Post-1832			−0.002	−0.004
			(0.002)	(0.004)
Constant	19.153***	0.081	19.084***	−0.073
	(0.169)	(0.274)	(0.194)	(0.314)
N	247	248	247	248
R^2	0.089	0.141	0.091	0.145
Adjusted R^2	0.082	0.134	0.080	0.134

Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

For my third and fourth models, given in Column 1-2 in Table 2, I included geographic controls such as average temperature and precipitation, distance to nearest coast, and distance to nearest port. I regressed both my dependent variables with these control variables.

Regression estimates from Table 2 for both log(GDP) and the log(NTL) reveal statistically significant interaction between treatment (Post-1832=1, Greek) and distance from the historical border, at 90% and 95% confidence levels. However, in the GDP model, the intercept is statistically significant in both the baseline model and the model with geographic controls. Therefore, I include the distance to the historical centers, old capitals, and major cities, namely Athens, Thessaloniki, and Istanbul. In addition, I also include TRI (terrain ruggedness index) as a control variable to account for topographic variance. OLS estimates controlled for distance to major cities adn TRI is reported in Table 3.

My regression estimates from Table 3 for both log(GDP) and log(NTL) reveal a statistically significant negative interaction between treatment (Post-1832=1, Greek)

Table 2: OLS Models with Geographic Controls

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.001	0.001
	(0.001)	(0.002)
Post-1832 (Greek = 1)	0.331	0.358
	(0.288)	(0.361)
Distance x Post-1832	0.004*	0.006**
	(0.002)	(0.002)
Distance to nearest coast (km)	0.006	0.016***
	(0.004)	(0.005)
Distance to nearest port (km)	−0.008**	−0.016***
	(0.004)	(0.005)
Average temperature 1971-2000	0.265***	0.425***
	(0.069)	(0.086)
Average precipitation 1971-2000	−0.031***	−0.037***
	(0.005)	(0.007)
Area site (km^2)	0.0002	−0.002***
	(0.0002)	(0.0003)
Constant	17.558***	−2.312
	(1.192)	(1.495)
N	247	248
R^2	0.362	0.641
Adjusted R^2	0.341	0.629

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

Table 3: OLS Models with Other Controls

	Dependent variable:			
	log(GDP) GDP (+ cities)	log(NTL) NTL (+ cities)	log(GDP) GDP (+ TRI)	log(NTL) NTL (+ TRI)
Distance to border (km)	0.016^{***}	0.027^{***}	0.014^{***}	0.024^{***}
	(0.004)	(0.004)	(0.003)	(0.004)
Post-1832 (Greek = 1)	0.308	0.081	0.292	0.024
	(0.310)	(0.368)	(0.296)	(0.329)
Distance x Post-1832	-0.017^{***}	-0.028^{***}	-0.015^{***}	-0.024^{***}
	(0.004)	(0.005)	(0.004)	(0.004)
Distance to Athens (km)	-0.012^{***}	-0.021^{***}	-0.013^{***}	-0.022^{***}
	(0.002)	(0.003)	(0.002)	(0.003)
Distance to Thessaloniki (km)	-0.007^{***}	-0.010^{***}	-0.006^{***}	-0.008^{***}
	(0.002)	(0.002)	(0.002)	(0.002)
Distance to Istanbul (km)	0.003^{*}	0.007^{***}	0.004^{**}	0.007^{***}
	(0.002)	(0.002)	(0.002)	(0.002)
Distance to nearest coast (km)	0.0003	0.004	−0.004	−0.003
	(0.005)	(0.005)	(0.004)	(0.005)
Distance to nearest port (km)	−0.002	−0.007	−0.002	-0.007^{*}
	(0.004)	(0.004)	(0.004)	(0.004)
Average temperature 1971–2000	0.273^{***}	0.399^{***}	0.089	0.077
	(0.070)	(0.084)	(0.077)	(0.085)
Average precipitation 1971–2000	0.018	0.035^{**}	0.021^{*}	0.042^{***}
	(0.013)	(0.016)	(0.013)	(0.014)
Area site (km^2)	0.0005^{**}	-0.001^{***}	0.001^{***}	-0.001^{***}
	(0.0002)	(0.0003)	(0.0002)	(0.0002)
TRI			-0.029^{***}	-0.051^{***}
			(0.006)	(0.006)
Constant	15.407^{***}	-5.417^{***}	18.377^{***}	−0.065
	(1.306)	(1.548)	(1.382)	(1.540)
N	247	248	247	248
R^2	0.437	0.714	0.490	0.774
Adjusted R^2	0.411	0.701	0.464	0.762

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

and distance from the historical border. While this finding supports my argument, the intercept is still significant for both models.

The OLS estimates reported in Table 3 show that the interaction term between Post-1832 status and distance to the historical border is statistically significant at conventional levels in both specifications. This interaction captures the differential spatial gradients on each side of the border: municipalities incorporated early into Greece experience almost no change in economic activity as distance increases, whereas municipalities that remained under Ottoman administration display a comparatively higher gradient. The predicted-values plots in Figure 7 for logged GDP and Figure 8 for logged night-lights visually illustrate this pattern. In both figures, the slope of the Greek municipalities is almost zero, while the slope for Ottoman municipalities is positive, and the predicted value increases as we move away from the cutpoint. This divergence in gradients is consistent with the argument that municipalities exposed to late-Ottoman administrative reforms—especially during the Tanzimat and constitutional periods—retained more stable and relatively better performing administrative and extractive capacity. In other words, although no level discontinuity exists at the 1832 border, the differential slopes suggest that the duration and depth of Ottoman institutional reforms shaped the internal spatial structure of local economic development.

For these findings, I conducted several robustness checks, including distance to highways and urban roads, elevation, and logged population density. My estimates are robust against distance to motorways and elevation, however, logged population density absorbs the treatment effect. This is expected, since population density and urbanization are primary mediating mechanisms of economic development. I suspect the effect of my treatment is absorbed by the mediating variable.

Figure 7: Predicted values of log(GDP) from Model 3 & 4 with full controls

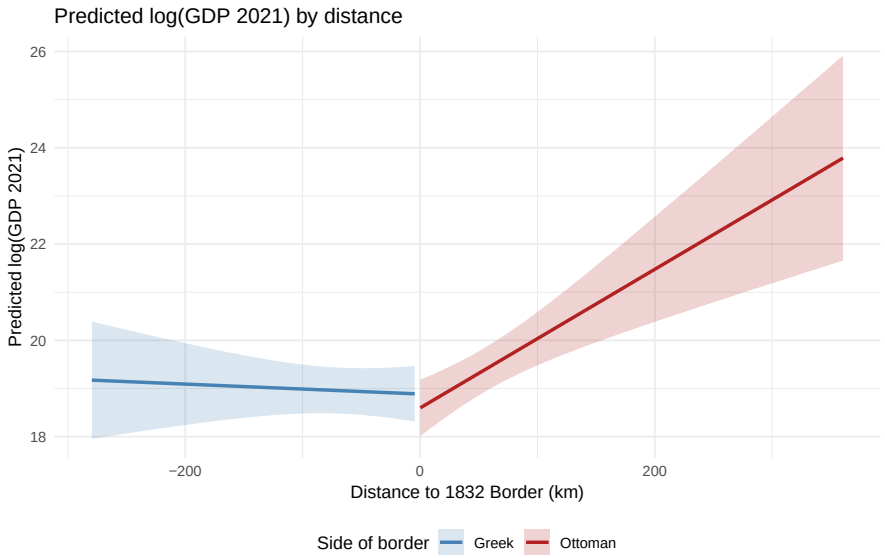
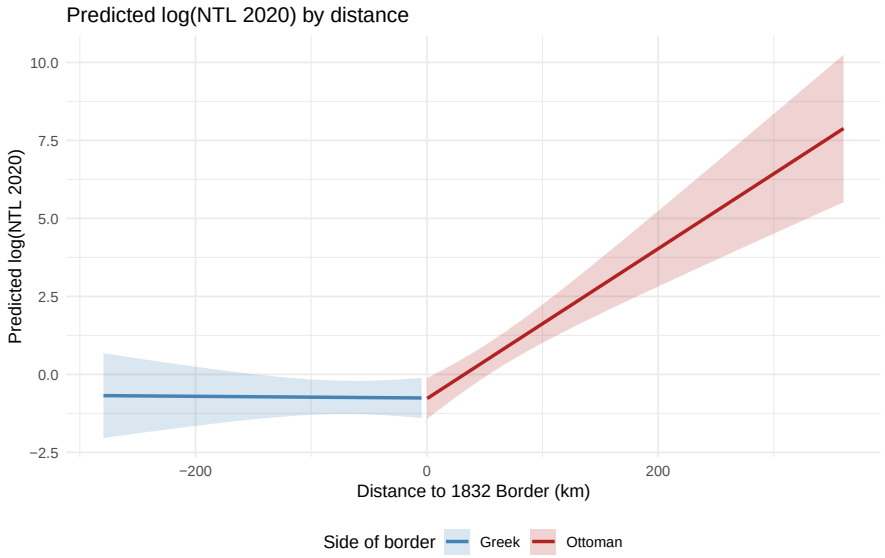


Figure 8: Predicted values of log(NTL) from Model 3 & 4 with full controls



RDD Estimates

The geographic RDD study design allows me to isolate the effect of institutional divergence, assuming that in the optimal bandwidth, observations in both sides of the cutpoint are identical in terms of potential confounders and concerns for heterogeneity.

Table 4: RDD OLS Models with Controls

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.006	0.013*
	(0.005)	(0.008)
Post-1832 (Greek = 1)	−0.196	0.207
	(0.378)	(0.380)
Distance x Post-1832	−0.023***	−0.011
	(0.006)	(0.011)
Distance to Athens (km)	−0.021***	−0.013
	(0.006)	(0.016)
Distance to Thessaloniki (km)	−0.030***	−0.014
	(0.008)	(0.020)
Distance to Istanbul (km)	0.017***	0.011
	(0.005)	(0.014)
Distance to nearest coast (km)	0.002	−0.004
	(0.006)	(0.008)
Distance to nearest port (km)	−0.010*	−0.008
	(0.005)	(0.006)
Average temperature 1971-2000	0.141	−0.032
	(0.117)	(0.131)
Average precipitation 1971-2000	0.044**	0.024
	(0.020)	(0.037)
Area site (km^2)	0.001**	−0.0003
	(0.0003)	(0.0003)
TRI	−0.022***	−0.033***
	(0.008)	(0.008)
Constant	14.202***	−1.876
	(2.458)	(4.777)
N	128	64
R^2	0.520	0.461
Adjusted R^2	0.470	0.334

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

While OLS models allow me to evaluate broad correlational patterns in the full sample, they fall short in isolating the institutional effect. Hence, I used subsamples

of my observations according to each model's IK-optimal bandwidth for a geographic RDD. As we can see in Table 4, for the IK-optimal bandwidth, the coefficient of the interaction term in my GDP model remains stable and significant at conventional levels. In contrast, in the NTL model, my interaction term is statistically insignificant, but this is mostly due to the low number of observations in the optimal bandwidth. However, in the full sample with controls, my interaction term in the NTL model is statistically significant.

Unfortunately, as Figure 9 and 10 shows, there is no sharp discontinuity around the border. Due to the lack of a statistically significant sharp discontinuity, we fail to reject the null hypotheses.

Figure 9: log(NTL) in 2020 (Optimal Bandwidth Sample)

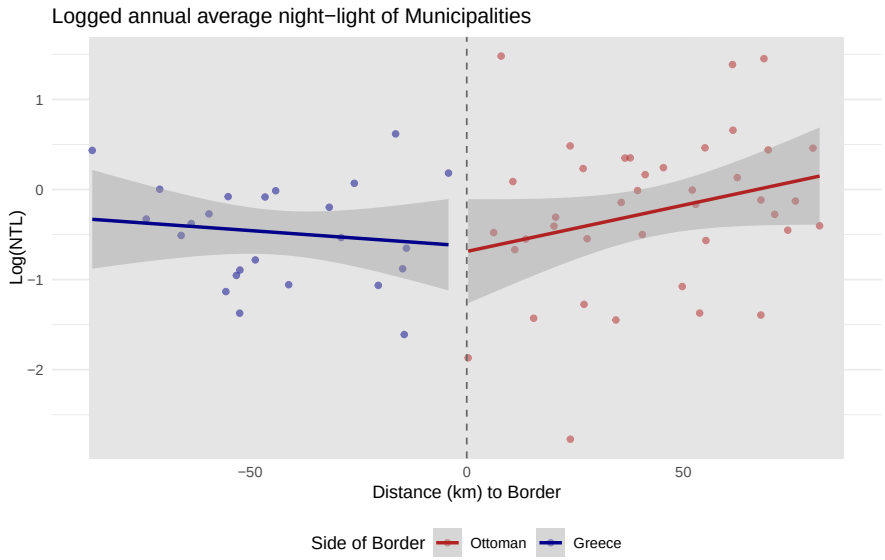
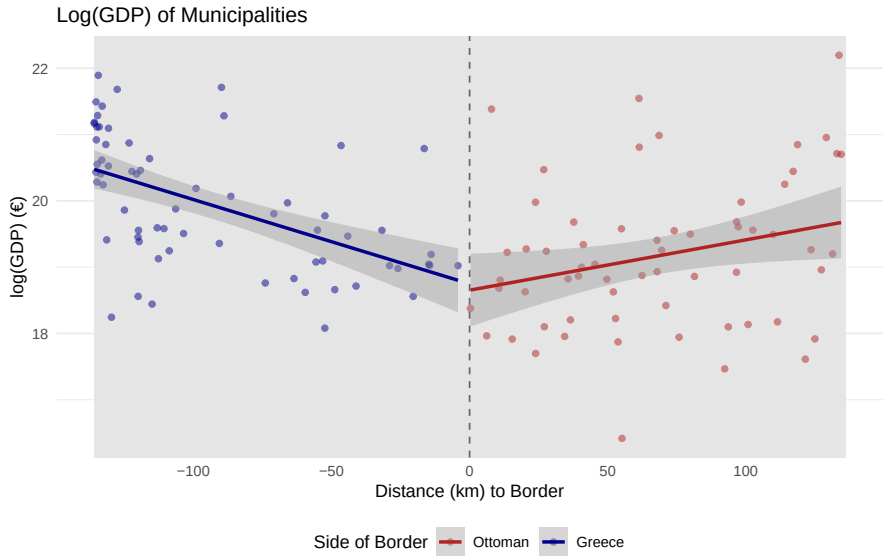


Figure 10: log(GDP) in 2021 (Optimal Bandwidth Sample)



The RDD estimates show no strong evidence of an immediate discontinuity at the 1832 border, suggesting that municipalities incorporated earlier into Greece do not exhibit clear level differences in contemporary economic outcomes once geography is controlled for—thereby weakening the classical narrative of a uniformly negative "Ottoman legacy". Instead, the most meaningful pattern appears in the interaction term: the Ottoman side displays a more favorable internal spatial gradient, with economic activity increasing as distance from the border increases, whereas Greek municipalities perform similarly across distance. This pattern is consistent with the institutional reforms implemented during the Tanzimat and constitutional periods, which aimed at administrative centralization, tax codification, cadastral modernization, and bureaucratic professionalization. This finding is in parallel to the longevity of Ottoman rule and higher exposure to the Tanzimat and later reforms, since the Ottoman Empire lost Thessaly in the late 19th century and Western Thrace in 1912. The longer these municipalities experienced these institutional reforms, economic activity increases, while ones that were not exposed stagnated. Municipalities that remained under Ottoman administration during

this period appear to show an increase in economic outcomes with distance, indicating deeper and more consistent institutional penetration compared to early Greek territories, which faced the challenge of building administrative structures largely from scratch. Thus, rather than revealing a sharp level-based treatment effect, the RDD results point to a structural difference in how space translates into development on each side of the historical boundary—a pattern that aligns closely with the theoretical expectation that institutional duration and reform depth, rather than mere imperial affiliation, shape long-run extractive capacity. These findings also shows the importance of 19th century institutional reforms in the Ottoman Empire, and should be taken into account while studying the total effect of Ottoman Imperial rule. Previous studies in the literature seem to overlook the impact of these reforms in the scope of a larger geography, however, when closely inspected the impact of these reforms is clear.

Discussion

The RDD estimates do not yield statistically significant discontinuities at the 1832 border, suggesting that the immediate local average treatment effect of Ottoman rule is limited once geographic and infrastructural controls are introduced. However, the OLS results—particularly those drawn from the full sample—provide consistent and statistically meaningful evidence that aligns with the expectation of a potential positive legacy of late nineteenth-century Ottoman institutional reforms. Contrary to the dominant findings in the existing literature (Popescu & Popa, 2022; Grosjean, 2011), which emphasize the adverse long-run effects of Ottoman rule on development, these estimates indicate that municipalities exposed to the Tanzimat and constitutional reforms may exhibit more favorable contemporary economic gradients. These relationships remain robust when controlling for elevation and proximity to major roads, reinforcing the plausibility of a persistent institutional mechanism. Only logged population density absorbs

the treatment effect entirely, suggesting that population concentration may function as a mediating channel rather than a confounder—potentially reflecting the demographic consequences of administrative stability under late Ottoman governance. Supplementary robustness estimates are reported in the appendix.

Nonetheless, several limitations must be acknowledged. Foremost, the models do not incorporate measures of temporal convergence. The dependent variables—GDP in 2021 and logged annual average night-light intensity in 2020—are contrasted against a treatment variable capturing institutional exposure nearly two centuries earlier. Without matching historical controls to historical treatment intensity, the risk remains that present-day covariates imperfectly proxy conditions near the time of the institutional shock. Future iterations of this analysis should incorporate historical datasets to more accurately capture pre-treatment or contemporaneous covariates, such as nineteenth-century population figures, administrative hierarchies, or localized tax registers. Additionally, instead of relying on modern travel times to major cities, an improved approach would incorporate historical cost-distance surfaces or mule-track routing networks to better approximate the geographic frictions and administrative reach of the period. These refinements will enable a more temporally coherent identification strategy and a more precise evaluation of whether Ottoman institutional reforms—rather than Ottoman rule per se—account for the observed spatial patterns in contemporary economic activity.

Conclusion

This paper examined the long-run economic consequences of Ottoman imperial rule by exploiting the first internationally recognized border between the Ottoman Empire and the Kingdom of Greece as a source of quasi-exogenous variation in institutional exposure. Using both OLS and geographic RDD specifications, I asked whether municipalities that remained under Ottoman control for a longer period—thus experiencing the Tanz-

imat and subsequent constitutional reforms—display systematically different patterns of contemporary economic development, proxied by logged GDP and logged night-light intensity.

The RDD estimates do not reveal a sharp discontinuity in levels of economic activity at the 1832 border once geography and infrastructure are controlled for. In that sense, the results provide little support for a simple "border as shock" interpretation of Ottoman rule, and they do not corroborate a uniformly negative legacy in this particular subnational context. However, the OLS models estimated on the full sample, combined with the RDD specifications that incorporate a rich set of geographic controls, point to a more subtle but substantively meaningful pattern: the slope of economic outcomes with respect to distance from the historical border differs systematically across sides. Municipalities on the Ottoman side exhibit a more favorable or stable spatial gradient, while economic activity declines more sharply as one moves inward on the Greek side. This differential gradient is robust to a bunch of geographic, climatic, and infrastructural controls, and remains visible even when restricting attention to municipalities within the IK-optimal bandwidth around the border.

Substantively, these findings are consistent with the view that the content and duration of institutional exposure matter as much as imperial affiliation itself. Rather than Ottoman rule being uniformly development-reducing, the results suggest that prolonged exposure to late nineteenth-century Ottoman reforms—centralization of administration, codification of tax and property regimes, and expansion of bureaucratic capacity—may have left a positive imprint on local extractive and administrative capacity. At the same time, the fact that logged population density absorbs the treatment effect in extended specifications underscores that any such institutional legacy likely operated through demographic and urbanization channels, rather than bypassing them.

The analysis also highlights several avenues for future research. First, a more temporally coherent design—linking historical covariates and pre-treatment outcomes to insti-

tutional exposure—would help distinguish between genuine reform effects and long-run selection into Ottoman versus Greek control. Second, extending this GRDD framework to other Ottoman frontiers, or to different dimensions of state capacity (such as taxation, schooling, or bureaucratic staffing), would help assess whether the mechanisms identified here generalize beyond this particular border. Finally, by foregrounding the importance of nineteenth-century reform episodes that much of the literature treats as epiphenomenal, this paper argues that evaluations of "Ottoman legacy" need to move beyond static, empire-wide characterizations and pay closer attention to when, where, and how imperial institutions actually changed. In doing so, it contributes to a broader research agenda that treats imperial rule as a heterogeneous bundle of institutions whose long-run effects depend on timing, depth, and local administrative penetration, rather than on imperial labels alone.

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Appendix

A1: Robustness Check A (Elevation)

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.016^{***}	0.028^{***}
	(0.004)	(0.004)
Post-1832 (Greek = 1)	0.311	0.080
	(0.308)	(0.363)
Distance x Post-1832	-0.018^{***}	-0.030^{***}
	(0.004)	(0.005)
Distance to Athens (km)	-0.014^{***}	-0.023^{***}
	(0.002)	(0.003)
Distance to Thessaloniki (km)	-0.008^{***}	-0.010^{***}
	(0.002)	(0.002)
Distance to Istanbul (km)	0.004^{**}	0.007^{***}
	(0.002)	(0.002)
Distance to nearest coast (km)	0.002	0.006
	(0.005)	(0.005)
Distance to nearest port (km)	-0.002	-0.007^{*}
	(0.004)	(0.004)
Average temperature 1971-2000	0.134	0.184
	(0.095)	(0.113)
Average precipitation 1971-2000	0.021	0.039^{**}
	(0.013)	(0.016)
Area site km^2	0.001^{***}	-0.001^{***}
	(0.0002)	(0.0003)
Mean elevation	-0.001^{**}	-0.001^{***}
	(0.0004)	(0.0005)
Constant	17.398^{***}	-2.326
	(1.594)	(1.886)
N	247	248
R^2	0.448	0.723
Adjusted R^2	0.420	0.709

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

A2: Robustness Check B (Elevation with TRI)

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.013^{***}	0.021^{***}
	(0.003)	(0.004)
Post-1832 (Greek = 1)	0.279	−0.008
	(0.293)	(0.318)
Distance x Post-1832	-0.013^{***}	-0.019^{***}
	(0.004)	(0.005)
Distance to Athens (km)	-0.011^{***}	-0.018^{***}
	(0.002)	(0.003)
Distance to Thessaloniki (km)	-0.005^{***}	-0.006^{***}
	(0.002)	(0.002)
Distance to Istanbul (km)	0.003^{*}	0.005^{***}
	(0.002)	(0.002)
Distance to nearest coast (km)	-0.008^{*}	-0.011^{**}
	(0.005)	(0.005)
Distance to nearest port (km)	−0.001	−0.005
	(0.004)	(0.004)
Average temperature 1971-2000	0.199^{**}	0.313^{***}
	(0.092)	(0.100)
Average precipitation 1971-2000	0.019	0.037^{***}
	(0.013)	(0.014)
Area site km^2	0.001^{**}	-0.001^{***}
	(0.0002)	(0.0002)
TRI	-0.043^{***}	-0.079^{***}
	(0.009)	(0.009)
Mean elevation	0.001^{**}	0.003^{***}
	(0.001)	(0.001)
Constant	16.973^{***}	-3.060^{*}
	(1.522)	(1.651)
N	247	248
R^2	0.500	0.789
Adjusted R^2	0.472	0.778

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

A3: Robustness Check C (Motorways)

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.008^{**}	0.014^{***}
	(0.003)	(0.003)
Post-1832 (Greek = 1)	0.094	−0.295
	(0.269)	(0.270)
Distance x Post-1832	-0.012^{***}	-0.018^{***}
	(0.004)	(0.004)
Distance to Athens (km)	-0.008^{***}	-0.014^{***}
	(0.002)	(0.002)
Distance to Thessaloniki (km)	-0.005^{***}	-0.007^{***}
	(0.002)	(0.002)
Distance to Istanbul (km)	0.001	0.002
	(0.002)	(0.002)
Distance to nearest coast (km)	−0.001	0.002
	(0.004)	(0.004)
Distance to nearest port (km)	−0.002	-0.009^{***}
	(0.003)	(0.003)
Distance to urban roads (km)	-0.197^{***}	-0.257^{***}
	(0.055)	(0.055)
Distance to major highways (km)	-0.037^{***}	-0.056^{***}
	(0.007)	(0.007)
Average temperature 1971-2000	0.135^{*}	0.160^{**}
	(0.069)	(0.070)
Average precipitation 1971-2000	0.014	0.031^{***}
	(0.011)	(0.011)
Area site km^2	0.001^{***}	-0.001^{***}
	(0.0002)	(0.0002)
TRI	-0.013^{**}	-0.024^{***}
	(0.006)	(0.006)
Constant	18.809^{***}	0.613
	(1.250)	(1.257)
N	247	248
R^2	0.588	0.851
Adjusted R^2	0.563	0.842

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

A4: Robustness Check D (log(population density))

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	−0.00000	0.005***
	(0.002)	(0.002)
Post-1832 (Greek = 1)	0.390**	0.116
	(0.191)	(0.159)
Distance x Post-1832	0.0005	−0.004
	(0.003)	(0.002)
Distance to Athens (km)	0.00005	−0.006***
	(0.002)	(0.001)
Distance to Thessaloniki (km)	−0.001	−0.002
	(0.001)	(0.001)
Distance to Istanbul (km)	0.0002	0.002**
	(0.001)	(0.001)
Distance to nearest coast (km)	−0.004	−0.002
	(0.003)	(0.002)
Distance to nearest port (km)	0.003	−0.002
	(0.002)	(0.002)
Average temperature 1971-2000	−0.015	−0.043
	(0.050)	(0.041)
Average precipitation 1971-2000	−0.003	0.011
	(0.008)	(0.007)
Area site km^2	0.002***	−0.00004
	(0.0001)	(0.0001)
TRI	−0.006	−0.019***
	(0.004)	(0.003)
Log(population density)	0.533***	0.678***
	(0.029)	(0.024)
Constant	16.977***	−1.824**
	(0.896)	(0.747)
N	247	248
R^2	0.788	0.947
Adjusted R^2	0.776	0.944

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.

A5: Robustness Check E (sanity check)

	Dependent variable:	
	log(GDP) GDP Model	log(NTL) NTL Model
Distance to Border (km)	0.002	−0.0002
	(0.001)	(0.002)
Post-1832 (Greek = 1)	0.918***	0.915*
	(0.321)	(0.520)
Distance x Post-1832	−0.002	−0.004
	(0.002)	(0.004)
Constant	19.084***	−0.073
	(0.194)	(0.314)
N	247	248
R^2	0.091	0.145
Adjusted R^2	0.080	0.134

Standard errors in parantheses. *p<0.1; **p<0.05; ***p<0.01
Two-tailed tests.